

**COLLEGE OF
CHARLESTON**
 DEPARTMENT OF MATHEMATICS

Abstract

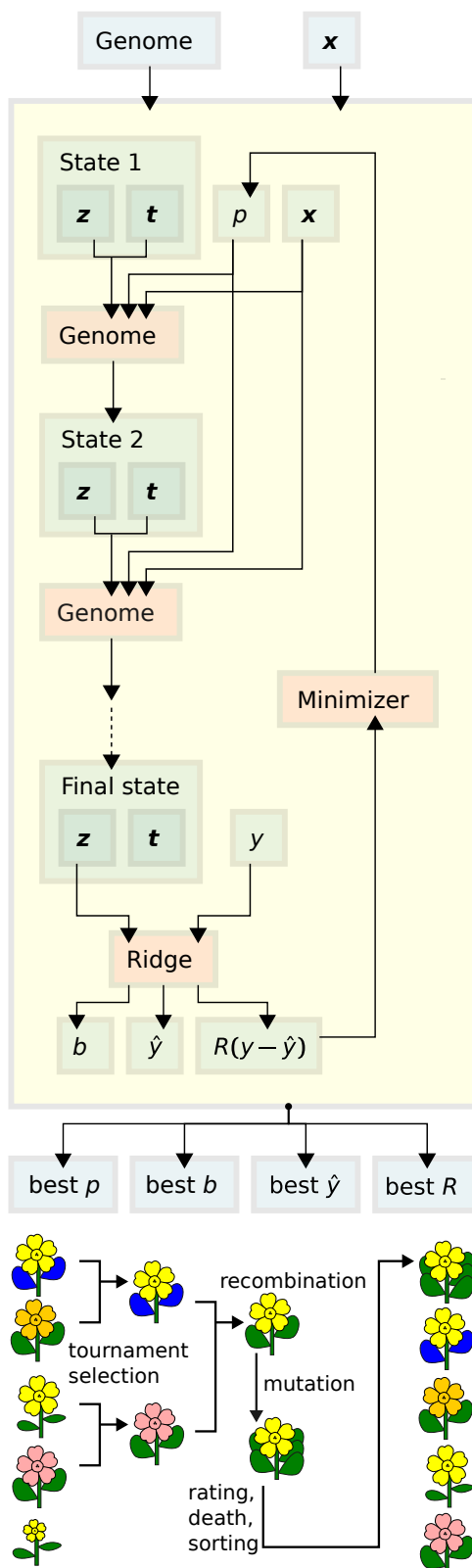
Machine learning

$$\underbrace{\sum_h |y_h - \hat{y}_h|^2}_{R(y - \hat{y})} + \text{regularization}$$

Contact Info

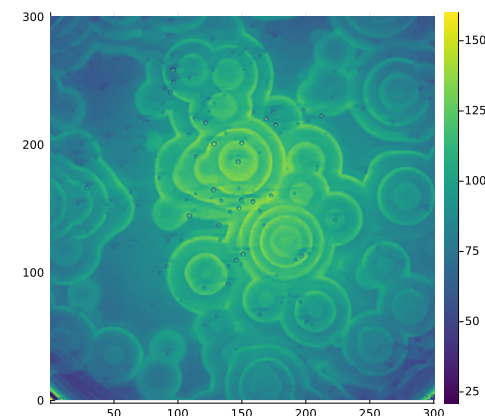
<http://github.com/wgm-applied-math/Jessamine.jl>

Overview



After solving for b 's and translating back to PDE variables, Jessamine recovers the PDE for u_t exactly, up to rounding errors. Similarly, v_t is recovered exactly.

The Belousov-Zhabotinsky reaction exhibits complex spatial dynamics. Can Jessamine infer meaningful PDEs from movie data?



BZ data from Alex Brummer and Whitney Kitchen